

Level 3



It's your turn to steer the spaceship

- You are given a list of space stations
- **Steer your spaceship to each space station**
- Space is **1-dimensional**
- For each space station:
 - Your spaceship starts at position 0
 - **Generate a movement sequence**
 - Your spaceship must reach the position of the station exactly
 - A time limit is given
 - The sequence must complete within the time limit
 - 7KH WLP H OLPLW LV YHU\ JHQHURXV \RX GRQTW QHHG D IDVW
- Allowed paces are explained on the next slide



Paces

- There are limitations on how your spaceship can move
- Each sequence must start and end with pace 0
- The slowest pace is 5 or -5
- The fastest pace is 1 or -1
- Your spaceship can only gradually step up the pace as well as step down

$-1 \Leftrightarrow -2 \Leftrightarrow -3 \Leftrightarrow -4 \Leftrightarrow -5 \Leftrightarrow 0 \Leftrightarrow 5 \Leftrightarrow 4 \Leftrightarrow 3 \Leftrightarrow 2 \Leftrightarrow 1$



Input

Name	Description	Example
N	Number of cases	3 2 25 5 39 -7 46
Space station and time limit (repeated N times)	The position of the space station and the time limit separated by a space	

Output

Name	Description	Example
Sequence (repeated N times)	A list of integers separated by spaces, each representing a pace	0 5 5 0 0 5 4 3 4 5 0 0 -5 -4 -3 -2 -3 -4 -5 0

Good Luck!



CLOUDELIGHT
CODING
CONTEST

#WeLoveSoftware
u too? cloudflight.io/career

